

Final Year Project

CB42

IoT model of in-car sensor and
telemetry system displayed on
screen and in the cloud

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Course - Higher Diploma in Computer Science

University - South East Technical University

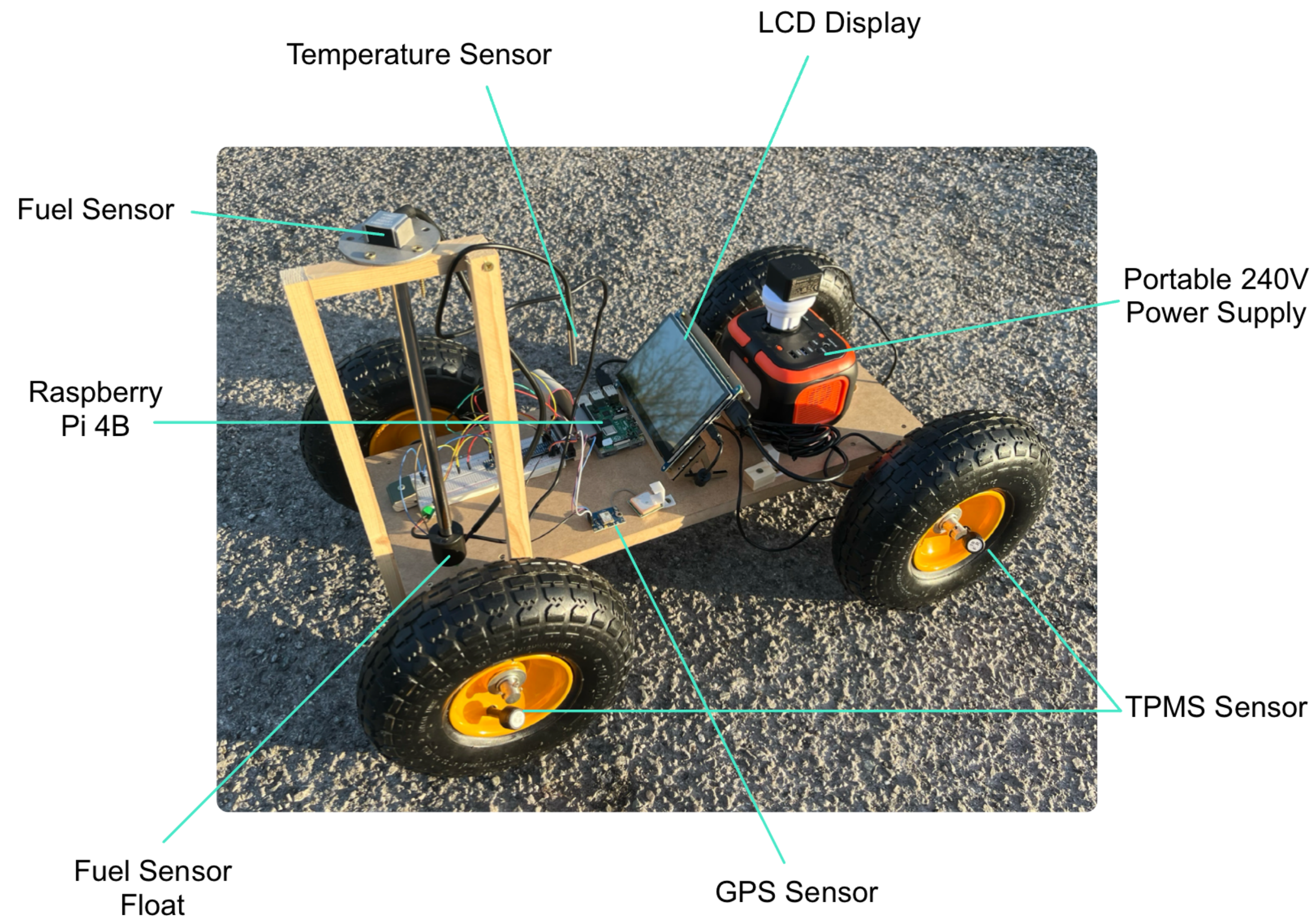
Outline

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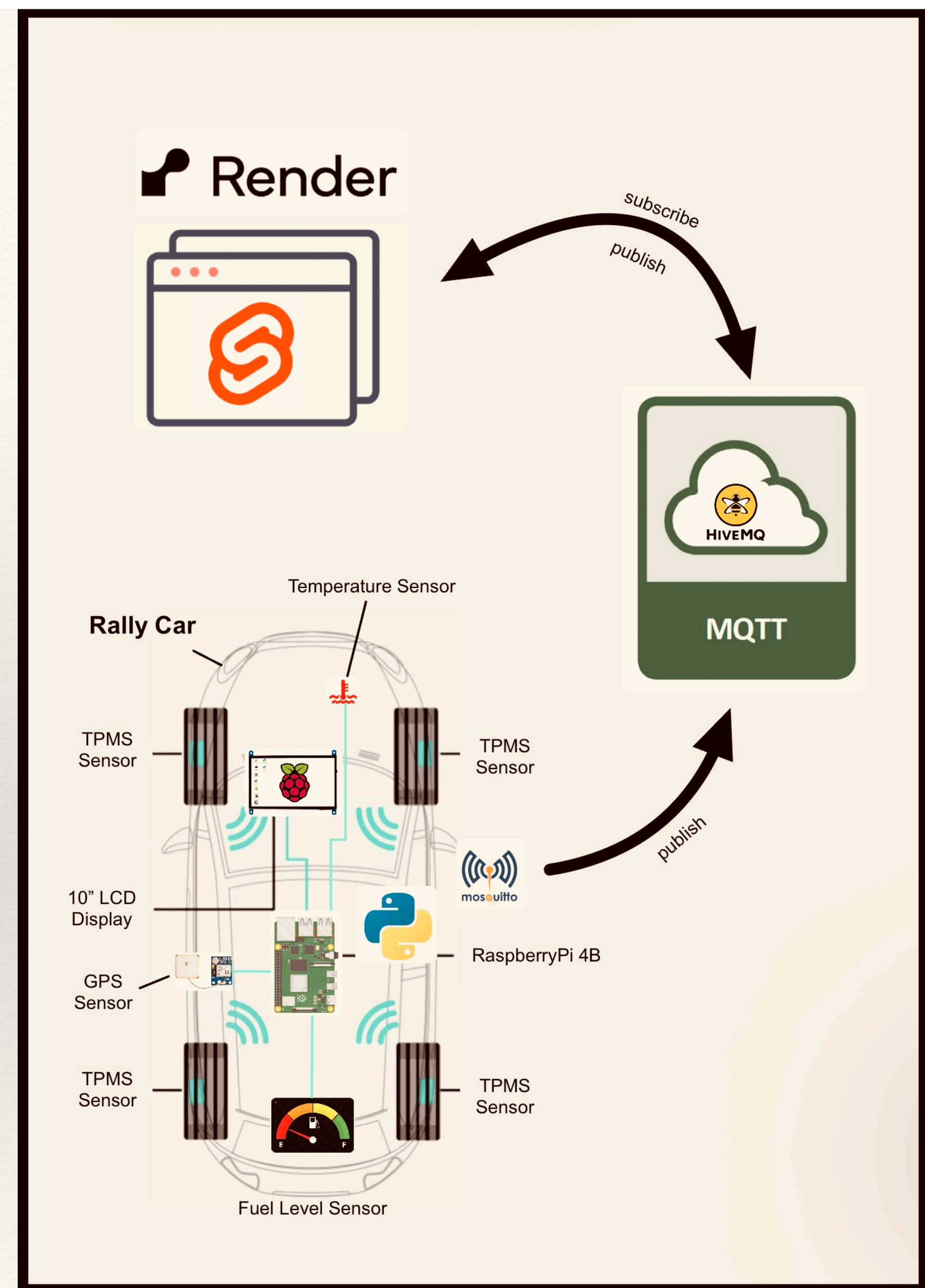
Background



Physical Prototype



System Diagram



Project Roadmap

1. Design overall architecture and order materials for prototype
2. Get pressure values from TPMS sensors printing to console
3. Get temperature readings from DS18B20 sensor printing to console
4. Get data from NEO-6M GPS sensor printing to console
5. Get percentage of fuel measure printing to console
6. Build physical prototype
7. Convert original sensor files to classes and build main script that imports these and publishes a single payload containing all data to local MQTT broker
8. Design in-car dashboard for Raspberry Pi display
9. Design web dashboard and set up cloud MQTT broker
10. Add warning for values that breach defined thresholds for both local and remote displays
11. Finalise code and report

Live Demo & Code Walkthrough

Pre-Recorded Backup: <https://www.youtube.com/watch?v=kxNXhQ9ZdNg>

Questions & Answers